

Experiment - 10a

Internetworking with routers in Cisco Packet Tracer Simulator.

Aim: Internetworking with routers in Cisco Packet Tracer Simulator, design a simple network with 1 router and 2 using Cisco Packet Tracer.

Procedure:

1) Configure router:

* Go to CLI → Enable → Config

* Set IP for Fast Ethernet 0/0 192.168.10.255.255.255.0 → no Shutdown

* Set IP for fast Ethernet 0/1 192.168.20.255.255.255.0 → no Shutdown

2) Configure PCs:

* PC0: IP 192.168.10.2, Gateway 192.168.10.1

* PC1: IP 192.168.20.2, Gateway 192.168.20.1

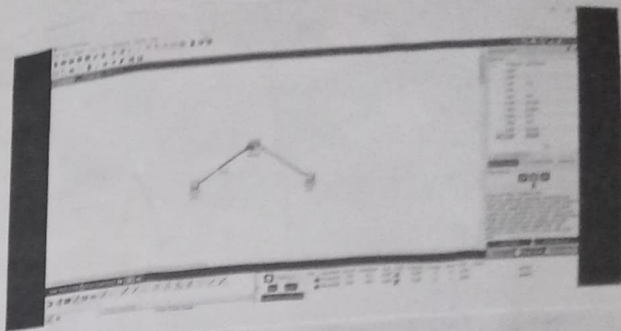
3) Connect Devices

* PC0 → Router Fast Ethernet 0/0 (copper straight through)

* PC1 → Router Fast Ethernet 0/1 (copper straight through)

4) Test Connectivity:

* Ping from PC0 to PC1 to verify network connection:



Experiment 10b

Aim:

Configure a network with a wireless router, DHCP Server and internet connection.

Procedure:

1) * add wireless router, PC, Laptop, cable modem, Internet cloud, Cisco.com server

* Connect using Copper and Coaxial cables as Required.

2) Configure wireless Router:

* Set SSID: Home Network

* Enable DHCP & Set DNS: 208.67.220.220

3) Configure Laptop & PC

* Laptop: Install wireless module → connect to Home network

* PC: Enable DHCP → Get IP automatically from router.

2) configure Cisco.com Server.

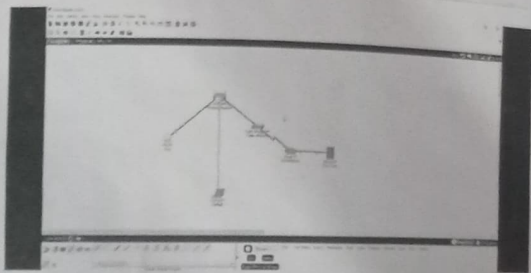
* Enable DHCP & DNS Services with
IP: 67.220.220.

* Set Pool and global gateway Settings.

Verify Connectivity:

* Refresh IP on PC (IP config / release
ipconfig / renew).

* Ping Cisco.com to test internet connect.



Student Observation

1) Write down the key features of configuring
wireless router & DHCP Server.

2) Wireless Router: Provides wifi - sets.
SSID connects devices.

DHCP Server: Automatically assigns
IP, gateway Subnet, DNS.

2) What is the Significance of DHCP Server
in internetworking?

A) * Simplifies IP Assignment & Avoids
conflicts.

* Ensures smooth network communication.

3) Design & configure an internetwork in your
lab using Switch, router & Ethernet
cables Draw & label.

4) * Setup, router → Switch → PCs (Ethernet
cables).

* Router: 192.168.1.1

* PC0: 192.168.1.2, Gateway: 192.168.

* PC1: 192.168.1.3, Gateway: 192.168.

* Test: Ping between PCs to verify
connection.

Result:

Internetworking with routers in
Cisco Packet Traces Simulator is done
Successfully.